

STANISLAV PITERIN

Smart Contract Security Auditor // EVM · DeFi · Oracles · RWA

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Smart contract security auditor working across the EVM DeFi stack: lending markets, vaults, AMMs and algorithmic stablecoins. Much of my recent work has centred on price-feed integrity, from oracle-priced venues for tokenised real-world assets to self-referential feeds and stale TWAP windows inside lending designs. I audit invariant-first: model the protocol, enumerate its invariants, then require a working Foundry PoC before recording any finding. I also build, which keeps the review grounded in what shipping the code actually costs.

► HEADLINE PROJECT

Yggdrasil — AI Advice Notary

Dec 2025 — present

An auditability model for AI dialogue state and engineering documentation. An MCP server fixes LLM dialogue state, so advice produced in conversation with a model is anchored as it is given and verifiable later against what was actually said.

Registry. Append-only Solidity contract — an independent anchor for records their own author could otherwise revise. Documents stay off-chain; `notarize(bytes32)` binds a SHA-256 hash to an author address and block timestamp, immutable by construction. `anchorRoot(bytes32, uint32)` amortises gas across a Merkle batch, membership proved off-chain.

Independent verification. A standalone Node CLI reads the chain directly, sharing no code with the desktop client, and reconciles filesystem, SQLite registry and on-chain record — `HASH_MISMATCH` / `MISSING_FILE`.

Stack. Solidity + Hardhat · Electron + React · Node CLI verifier · MCP server · 60+ tests. Detects tampering post factum, does not prevent it.

► SECURITY AUDITS

Independent Smart Contract Security Researcher

Apr 2026 — present

- **METRIC** — oracle-priced DEX for tokenised real-world assets (equities, FX, commodities). Private engagement, ongoing; running the public contest in parallel. Focus: oracle staleness, off-hours price manipulation, RWA-specific vulnerability classes.
- **ICFT** (ongoing) — commissioned pre-audit security review of a lending protocol, nine modules in scope: lending pool, interest-rate model, oracle layer, liquidation engine, DEX integration, access control and vesting. Architecture review surfaced a self-referential price feed sourced from the protocol's own thin pool, a missing bad-debt backstop, and a keeper design crowding out external liquidators. Invariant and economic stress scenarios running in Foundry.
- **Pinto Protocol** (Jun 2026) — algorithmic stablecoin on Base, Beanstalk-derived. Scope: 31 Diamond (EIP-2535) facets, 386 external methods. Reconstructed the economic model, mapped Diamond storage, built an invariant inventory, and ran fork-delta analysis against 20 prior audits to isolate new surface.
- **USDaVault** (Jun 2026) — solo audit of an ERC-4626 vault on Uniswap v4 concentrated liquidity. Architecture review, invariant derivation, line-by-line manual pass, Foundry PoC for every confirmed finding. Report: github.com/WhyUSayMyName/USDaVault-AUDIT
- **Contest portfolio** — Barebones MultipliVault (May 2026), Tokenizelt, SuperEarn (Apr 2026), MultiSigTimelock. Sherlock · Cantina · CodeHawks · Cyfrin.

► TOOLING

skill-proofhound-audit

Self-built audit automation: cross-function dependency graphs over the scope, vulnerability hypotheses from three independent sources, and a mandatory reproducible PoC before a hypothesis becomes a finding. Cuts false positives and recon time on large scopes.
github.com/WhyUSayMyName/skill-proofhound-audit

► DOMAIN EDGE

Mining engineering: Lane's cutoff grade theory, stripping economics, open-pit operations. Applicable to commodity-backed and real-world-asset collateral, where failure modes live in the price model as much as in the code.

► EDUCATION

Saint Petersburg Mining University (Empress Catherine II)
Specialist degree, Mining Engineering — Open Pit Mining · 3rd year · expected 2029
★ Winner, XXI International Forum-Contest “Topical Issues of Subsoil Use”
Statemind CTF certificate · BlockFirst: Smart Contract Auditor, Solidity Developer

► PUBLICATIONS

Smart Contracts as an Instrument of Digital Transformation in the Mining Industry
Socio-Economic and Ecological Problems of the Mining Industry, 2025
Distributed Ledger for Verifying the Integrity of Digital Data in Mining
Topical Issues of Subsoil Use, 2026 · both co-authored with E. V. Loginov · Scopus-indexed

► SKILLS

Languages & frameworks	Solidity, Foundry (forge / anvil / cast), Hardhat, Python (differential testing, modelling), TypeScript / Node, React, Electron, Git
Analysis	Slither, Aderyn, Surya, custom Foundry invariant and fuzz harnesses, dependency-graph recon tooling
Methodology	Invariant derivation and cataloguing, systematic missing-check and modifier audit, fork-delta analysis, PoC development, severity calibration, scope and dedup analysis
Vulnerability classes	Reentrancy (classic / cross-function / read-only), oracle and price manipulation (flash-loan, TWAP / EMA), access-control violations, share inflation / first depositor, accounting and rounding errors, signature replay (EIP-712 / permit), proxy storage collisions, MEV and front-running, DoS and griefing, unchecked external calls
Standards & mechanisms	ERC-20 / 4626 / 1155 / 721, Diamond (EIP-2535), upgradeable proxies (UUPS / Transparent / ERC-1967), AMM designs (constant product, concentrated liquidity, stableswap), lending and interest-rate models, staking and LST / LRT accounting, cross-chain messaging and bridge security